

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	19 July 2026
Team ID	
Project Name	Nutrition Assistant
Team Members	Kulasri Durga Deepika Bora Polimera Girija(Team lead)
Maximum Marks	4 Marks

Technical Architecture

NutriPulse Solution Architecture

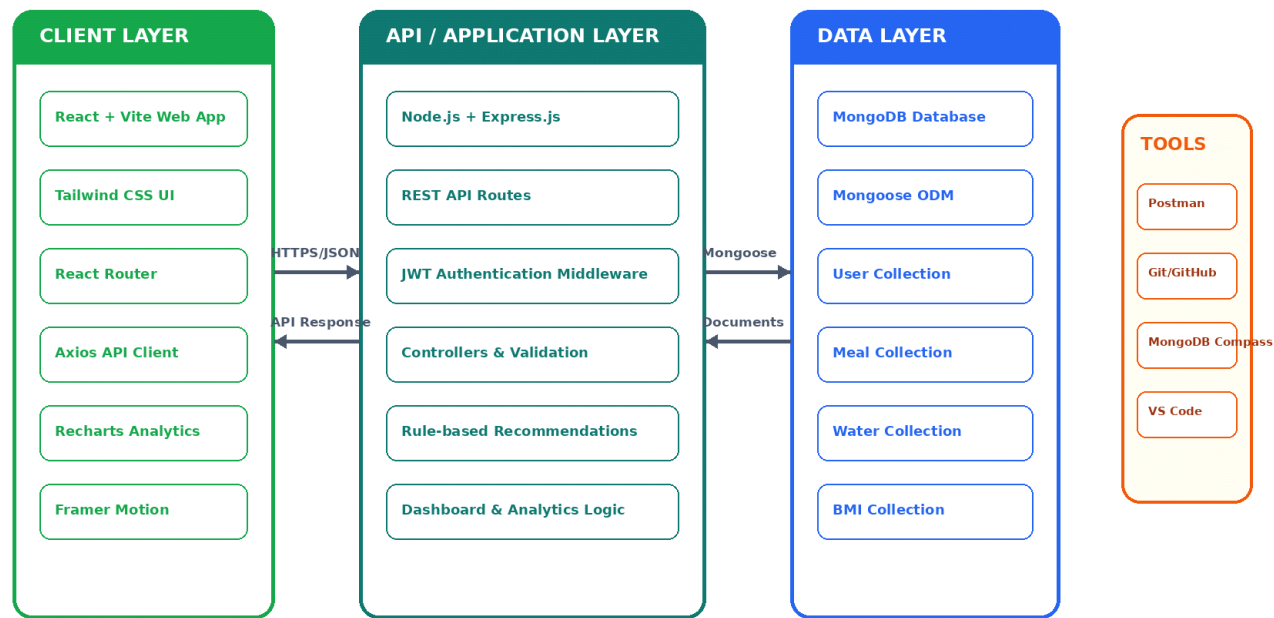


Table 1: Components & Technologies

S.No.	Component	Description	Technology
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1	User Interface	Responsive web interface and user interaction	React, Vite, Tailwind CSS, React Icons, Framer Motion
2	Routing	Public and protected page navigation	React Router DOM
3	API Communication	Client-server HTTP communication and JWT header injection	Axios
4	Application Server	REST APIs and middleware	Node.js, Express.js
5	Authentication	Token-based authentication and password protection	JSON Web Token, bcryptjs
6	Validation & Business Logic	Request validation, health calculations and recommendations	JavaScript, Express controllers
7	Database	Persistent storage for users and health records	MongoDB
8	Object Data Modeling	Schemas, validation, indexes and queries	Mongoose
9	Analytics & Visualization	Weekly charts and dashboard visualizations	Recharts
10	Testing Tools	Manual API and database verification	Postman, MongoDB Compass
11	Version Control	Source code management and collaboration	Git, GitHub
12	Development Environment	Code editing and local execution	VS Code, Nodemon, npm
13	Deployment (Recommended)	Public hosting for final demonstration	Vercel/Netlify, Render, MongoDB Atlas

Table 2: Application Characteristics

S.No.	Characteristic	Description	Technology / Approach
1	Open-source Frameworks	Built using widely adopted open-source web frameworks and libraries.	React, Express, Mongoose, Tailwind CSS, Recharts
2	Security Implementation	Passwords are hashed; APIs use JWT; routes are protected; CORS and environment variables are configured.	bcryptjs, JWT, Express middleware, dotenv
3	Scalable Architecture	Frontend, API and data layers are separated; feature modules use independent routes, controllers and models.	Three-tier modular architecture
4	Availability	Local services run independently; cloud deployment can provide continuous public access.	Render/Vercel/MongoDB Atlas – recommended

5	Performance	Parallel dashboard queries, MongoDB indexes and single analytics endpoint reduce unnecessary requests.	Promise.all, Mongoose indexes, API aggregation
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